Machine Learning Engineer | Assessment
Abdallah Aljabrah

Validation approach

A random train/test split was intentionally avoided.

Because the final prediction dataset represents future dates ,the validation strategy was designed to simulate **future period forecasting**.

The approach was : doing a chronological split to preserve the temporal structure of the data.

Training:
January /1/ 2025 → August /31/ 2025

Validation:
September /1/ 2025 → October /31/ 2025

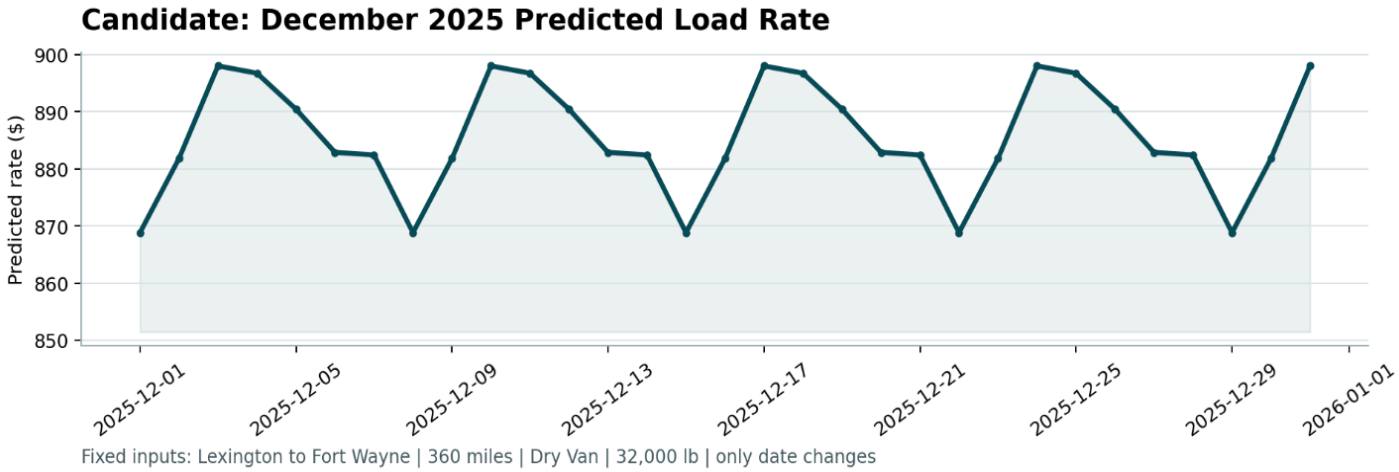
This resulted in:

- **38,477 training rows**
- **9,523 validation rows**

The validation represented almost 20% of the labeled dataset

Fixed December Prediction Chart

The following chart was generated using the provided score.py script after producing valid December predictions.



Note : the next pages contains a detailed report of what I did through the assessment , I truly thank for this opportunity to work on this assessment, the freight-rate dataset was something new to me and I learned a lot of things by exploring it and understanding it .

Data Exploration

The training dataset contained 14 original columns:

- load_id
- pickup
- delivery
- pickup and delivery coordinates
- distance
- equipment
- weight
- date
- market_index
- quote_signal
- posted_rate

The prediction target was identified as: posted_rate

The dataset contained **64 pickup cities**, **64 delivery cities**, and approximately **4,014 unique origin destination routes**.

Distance had the strongest direct linear relationship with posted_rate, with a correlation of approximately **0.91**, indicating that trip distance is a major pricing driver.

However, route, equipment type, and temporal information also showed useful pricing patterns.

Data Quality Issues

Several data quality issues were identified during exploratory analysis.

Missing weight values

The weight column contained:

- **300 missing values**
- **292 negative values**

The absolute magnitudes of negative weights closely matched the distribution of valid positive weights.

For example:

- negative absolute median: approximately **31,822 lb**
- positive median: approximately **31,494 lb**

This suggested that the negative values were likely sign errors rather than unrelated invalid values.

Negative weights were therefore corrected using absolute value :

```
WEIGHT = ABS(WEIGHT)
```

Missing weights were filled using the median weight from the labeled development data.

Equipment specific weight distributions were also investigated. Median weights were very similar across Dry Van, Flatbed, and Reefer loads, so a global median was considered sufficient.

Missing market_index

The market_index column contained **374 missing values**.

Exploration showed that market_index varied substantially over time. Monthly averages ranged approximately from:

- **0.89** in weaker periods
- to around **1.30** in stronger periods.

Because of this temporal behavior, replacing missing values using one global median as I did with **Missing weight values** would discard useful time information.

Instead, missing market_index values were filled using the median value from other loads occurring on the same date.

Daily groups contained roughly 125–188 valid observations, while within-day standard deviations were generally only around **0.02–0.03**, making same day median imputation a suitable approach.

A global development set median was retained only as a fallback.

Feature Engineering

Several additional features were derived from columns available at prediction time.

Date features

From date:

- month
- day_of_week_num

were extracted.

Route feature

A combined origin-destination category was created:

pickup -> delivery

For example:

Lexington -> Atlanta

Route level analysis showed noticeable variation in rate-per-mile across different origin destination pairs, suggesting that geography contains information beyond distance alone.

Nearly all validation routes were also observed in the training period.

In the October validation set:

- 2,598 unique routes appeared,
- only 6 routes were unseen during training

Validation Strategy

A random train/test split was intentionally avoided.

Because the final prediction dataset represents future dates, the validation strategy was designed to simulate future period forecasting.

Primary chronological split

Training:

January 1, 2025 → September 30, 2025

Validation:

October 1, 2025 → October 31, 2025

This resulted in:

- **43,147 training rows**
- **4,853 validation rows**

The validation set represented approximately 10.1% of the development dataset.

This setup evaluates whether a model trained on historical loads can generalize to a later unseen month.

Secondary two month backtest

A second validation experiment was performed to better match the final prediction horizon.

Training:

January 1, 2025 → August 31, 2025

Validation:

September 1, 2025 → October 31, 2025

This resulted in:

- **38,477 training rows**
- **9,523 validation rows**

The final unseen prediction set spans November and December, so this two month backtest provided a closer approximation of the final deployment setting.

Model Experiments

Three main approaches were evaluated.

1- **Ridge Regression baseline**

A Ridge regression model was used as a simple linear baseline.

Categorical features were converted using one-hot encoding, while numeric features were passed through directly.

Baseline features included:

- pickup
- delivery
- equipment
- route
- distance
- weight
- month
- day of week

October validation performance:

Metric	**Ridge**
MAE	**$234.36**
RMSE	**$670.52**
R^2	**0.808**
MAPE	**15.20%**

The model performed reasonably for typical loads but strongly underpredicted rare high price loads.

2- CatBoost — common feature model

CatBoost was selected because it can directly handle categorical features and nonlinear interactions without requiring one-hot encoding.

The common feature model used only features also available in the supplied fixed December input file:

- pickup
- delivery
- distance
- equipment
- weight
- month
- day of week
- route

October validation results:

Metric	Common CatBoost
MAE	**$122.06**
RMSE	**$647.84**
R^2	**0.820**
MAPE	**6.31%**

This represented a big improvement over Ridge, particularly in MAE and MAPE.

CatBoost — rich-feature model

A second CatBoost model added all additional fields available in the full validation dataset:

- pickup/delivery coordinates
- market_index
- quote_signal

along with the common features.

October validation results:

Metric	Rich CatBoost
MAE	**$117.98**
RMSE	**$646.95**
R^2	**0.821**
MAPE	**5.91%**

The richer feature set provided a consistent improvement.

Final Model Selection

The **Rich CatBoost model** was selected for the final 12,000 load predictions because:

- it performed best in both chronological validation experiments,
- it reduced MAE and MAPE compared with the common model,
- and all of its required features were available in validation.csv

Model	MAE	RMSE	R^2	MAPE
Common CatBoost	$122.60	$632.65	0.8281	6.11%
Rich CatBoost	**$108.61**	**$632.11**	**0.8284**	**5.17%**

Note : A separate Common CatBoost model was used for the supplied December fixed input scenario because that file does not contain:

- **market_index**
- **quote_signal**
- **coordinate features**